

1 Computer and Its Components

A **computer** is an electronic device that processes data to produce meaningful information.

It performs four main functions: **Input → Process → Output → Storage**.

Block Diagram of a Computer

1. **Input Unit:** Devices used to enter data (e.g., Keyboard, Mouse, Scanner).
 2. **Central Processing Unit (CPU):**
 - **Arithmetic Logic Unit (ALU):** Performs calculations and logic operations.
 - **Control Unit (CU):** Directs operations of all components.
 - **Registers:** High-speed storage locations within the CPU.
 3. **Memory/Storage Unit:**
 - **Primary Memory (RAM, ROM)** – Stores data temporarily or permanently for quick access.
 - **Secondary Memory (Hard Disk, SSD, CD/DVD)** – Long-term data storage.
 4. **Output Unit:** Devices that display processed information (e.g., Monitor, Printer).
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2 Input and Output Devices

- **Input Devices:** Help to feed data into the computer.
Examples: Keyboard, Mouse, Scanner, Microphone, Webcam.
 - **Output Devices:** Display or produce information.
Examples: Monitor, Printer, Speakers, Plotter.
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3 Memory Hierarchy

The **memory hierarchy** shows levels of storage speed and capacity.

	Level Type	Example	Speed	Capacity
1	Registers	CPU Registers	Fastest	Very Small
2	Cache Memory	L1, L2 Cache	Very Fast	Small
3	Primary Memory	RAM, ROM	Fast	Moderate
4	Secondary Memory	Hard Disk, SSD	Slower	Large

👉 **Concept:** Faster memory is more expensive but smaller in size.

4 Concept of Hardware and Software

- **Hardware:** Physical components of a computer (CPU, Monitor, Keyboard).
- **Software:** Set of instructions that tell hardware what to do.

Types of Software

1. **System Software:** Controls hardware and runs the computer (e.g., Windows, Linux).
 2. **Application Software:** Performs specific tasks (e.g., MS Word, Photoshop).
 3. **Utility Software:** Helps maintain the system (e.g., Antivirus, Backup tools).
 4. **Open Source Software:** Free to use and modify (e.g., LibreOffice).
 5. **Freeware:** Free to use but cannot be modified (e.g., Adobe Reader).
 6. **Proprietary Software:** Licensed and paid (e.g., Microsoft Office).
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5 Programming Language, Compiler, Interpreter, Translator

- **Programming Language:** A way to write instructions for computers (e.g., C, Python, Java).
- **Compiler:** Translates the entire program into machine code at once.
- **Interpreter:** Translates and executes code line by line.
- **Translator:** A general term for any program that converts high-level code into machine code.

6 Concept of Computing and Units of Memory

- **Computing:** The process of using computers to perform calculations or process data.
 - **Units of Memory:**
 - 1 Bit = Smallest unit (0 or 1)
 - 1 Byte = 8 Bits
 - 1 KB = 1024 Bytes
 - 1 MB = 1024 KB
 - 1 GB = 1024 MB
 - 1 TB = 1024 GB
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7 Operating System (OS) and Its Types

An **Operating System** is system software that manages hardware and software resources.

Functions of Operating System:

- Manages files and directories.
- Allocates memory to programs.
- Handles input/output operations.
- Provides user interface (GUI/CLI).
- Manages system security and multitasking.

Types of Operating Systems:

1. **Single-user OS:** One user at a time (MS-DOS).
2. **Multi-user OS:** Multiple users simultaneously (UNIX).
3. **Multitasking OS:** Runs several programs at once (Windows, macOS).
4. **Network OS:** Manages networked computers (Windows Server, Linux).
5. **Real-time OS:** Used in embedded systems (Automated machines).

8 Networking Concepts (LAN, MAN, WAN)

Networking connects computers to share data and resources.

Type	Full Form	Area Covered	Example
LAN	Local Area Network	Small area (Office, Lab)	Wi-Fi in a school
MAN	Metropolitan Area Network	City-wide	Cable TV network
WAN	Wide Area Network	Global	Internet

- **Internet:** The largest WAN connecting millions of computers worldwide.
- **Common Networking Devices:** Router, Switch, Modem, Hub.

UNIT II – Operating Systems & Word Processing (Practical)

1. Operating Systems (OS)

Definition

An **Operating System (OS)** is system software that manages computer hardware, software resources, and provides common services for computer programs.

A. Graphical User Interface (GUI) Basics

A **GUI** allows users to interact with computers using **icons, windows, and menus**, instead of typing text commands.

Examples:

- **Windows OS**
- **Linux (Ubuntu, Fedora)**

Key GUI Elements:

Element	Description
Desktop	The main screen that holds icons, shortcuts, and files.
Taskbar	Displays open applications and the Start menu.
Start Menu	Central point to access programs, files, and system settings.
Icons	Small images that represent applications, files, or shortcuts.
Windows	Rectangular areas on screen for running applications.

B. Basic Operations

Students learn how to:

- **Start and quit programs** from the Start Menu.
 - **Add or remove desktop icons.**
 - **Switch between multiple applications.**
 - **Work with multiple windows (resize, minimize, maximize, close).**
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C. File and Folder Management

Managing files efficiently is a core skill.

Common File Operations:

Task	Description
Create	Make a new file or folder.
Delete	Remove unnecessary files or folders.
Rename	Change the name of a file or folder.
Move	Transfer a file to another location.
Copy	Duplicate a file or folder.

Task	Description
Open	Access a file or folder to use it.

 **Practical Exercise:**
Create folders like DTP_Practice → Word_Files → Assignments.
Move and rename files inside using Windows Explorer.

D. Control Panel & Settings

Used for **system configuration** and customization.

Examples of Control Panel Functions:

- Changing **desktop background** or **theme**
 - Managing **user accounts**
 - Setting **date and time**
 - Installing/uninstalling programs
 - Configuring **hardware devices** (keyboard, mouse, printer)
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E. Accessories and File Management Tools

Windows includes small tools like:

- **Calculator**
- **Notepad**
- **WordPad**
- **Snipping Tool**
- **File Explorer**

These tools help in daily productivity and system navigation.

2. Word Processing (MS Word or OpenOffice Writer)

Definition

A **Word Processor** is an application used to create, edit, format, and print text-based documents like letters, reports, and resumes.

Examples:

- Microsoft Word
- LibreOffice Writer

A. Basic Operations

- **Opening/Creating Documents** – Start a new file or open an existing one.
- **Saving** – Store your document in the desired folder (.docx or .odt).
 - **Cut, Copy, Paste** – Move or duplicate text or images.
 - **Print & Print Preview** – Check layout before printing.

B. Text Formatting

Enhancing the appearance of text for clarity and presentation.

Feature	Description
Font Type & Size	Choose style and size of text.
Bold, Italic, Underline	Highlight text.
Alignment	Align text (Left, Center, Right, Justify).
Line Spacing	Adjust vertical spacing between lines.
Bullets & Numbering	Organize lists clearly.
Borders & Shading	Add emphasis or style to sections.

 Exercise:

Create a document with a heading, paragraph, and bullet points. Apply bold and different font colors.

C. Page Setup & Layout

- **Margins, Orientation, and Size** — Adjust page design.
 - **Header and Footer** — Add consistent information (e.g., page numbers, titles).
 - **Columns** — Arrange text in newspaper style.
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D. Tools for Editing and Proofing

Tool	Purpose
Spell Check	Automatically checks spelling errors.
Grammar Check	Identifies grammatical mistakes.
AutoCorrect	Fixes common typing errors.
Find and Replace	Quickly change repeated words or phrases.

E. Inserting Elements

Students learn to insert and manage:

- **Symbols and Equations**
- **Clip Art / Smart Art**
- **Pictures / Shapes**
 - **Tables**
 - **Hyperlinks**
 - **Bookmarks**
- **Cross-references**

Exercise:

Create a formatted letter that includes a table, picture, and hyperlink.

F. Mail Merge

Mail Merge combines a main document with a data source (like a list of names) to create personalized documents automatically.

Example:

Generating customized letters for multiple recipients.

Steps:

1. Create a main letter.
2. Prepare a data source (e.g., Excel file with names and addresses).
3. Link both and preview merged results.

G. Labels & Envelopes

Students learn how to:

- Print address **labels** for bulk mail.
- Create **envelopes** with return and delivery addresses.

UNIT III – Spreadsheet Applications (Practical)



1. Introduction to Spreadsheet

Definition

A **Spreadsheet** is an application used to store, organize, and analyze data in rows and columns using formulas and functions.

Examples:

- Microsoft Excel
- LibreOffice Calc
- Google Sheets

File Extensions:

.xls, .xlsx, .ods



2. Spreadsheet Interface Overview

Key Components:

Component	Description
Workbook	A complete Excel file containing one or more worksheets.
Worksheet	A single sheet within a workbook where data is entered.
Cell	Intersection of a row and column (e.g., A1).
Cell Address	Unique reference of each cell (like B5).
Range	A group of adjacent cells (e.g., A1:A10).
Formula Bar	Shows or edits the contents of the active cell.
Name Box	Displays the address of the selected cell.

Example:

When you open Excel, each new file is a **workbook** containing multiple **worksheets** named “Sheet1,” “Sheet2,” etc.

3. Basic Operations in Spreadsheet

Students learn how to:

- Open, create, and save **workbooks** and **worksheets**.
- Enter **text, numbers, and formulas** into cells.
- Use **Cut, Copy, Paste**, and **Fill Series** (for dates or numbers).
 - Insert or delete **rows and columns**.
 - Adjust **column width** and **row height**.
- Use **Print and Print Preview** for page setup.

Exercise Example:

Create a worksheet for monthly expenses with columns like *Item Name*, *Quantity*, *Price*, *Total*.

4. Formatting Cells

Cell Formatting Features:

Feature	Purpose
Font & Size	Change the appearance of text/numbers.
Cell Color & Borders	Highlight important data.
Text Alignment	Left, center, or right-align data.
Merge Cells	Combine multiple cells into one.
Text Wrapping	Display text on multiple lines within a cell.
Number Formatting	Format as Text, Currency, Date, or Percentage.

Exercise:

Create a student marksheet table and format it using borders, colors, and merged title cells.

5. Formulas and Functions

A. Formulas

Formulas are **user-defined equations** used to perform calculations on data.

Example:

=A2+B2 → Adds the contents of cells A2 and B2.

=A1*B1 → Multiplies values in A1 and B1.

B. Functions

Predefined formulas that perform specific calculations automatically.

Category	Example Functions	Purpose
Mathematical	SUM(), AVERAGE()	Basic calculations
Logical	IF(), AND(), OR(), NOT()	Conditional testing
Text	LEFT(), RIGHT(), LEN()	Manipulate text
Date	TODAY(), NOW()	Work with dates and times

Nested IF Example:

=IF(A2>=50,"Pass","Fail")

→ Displays “Pass” if marks ≥ 50 , otherwise “Fail”.

💡 Exercise:

Create a salary sheet that calculates Total Salary, Tax Deduction, and Net Salary using formulas.

🔍 6. Data Management: Sorting, Filtering, and Validation

A. Sorting

Arranges data in **ascending or descending order** (e.g., A–Z, smallest–largest).

Used to organize records systematically.

B. Filtering

Displays only the data that meets specific conditions.

Example: Show only students who scored above 80 marks.

C. Advanced Filter

Allows more complex conditions (multiple criteria at once).

D. Data Validation

Restricts data entry to specific types (e.g., numbers only, date range, dropdown list).

💡 Exercise:

Create a data sheet of students and use filters to show only those from a particular class.

📊 7. Charts and Graphs

Definition

Charts represent **data visually** for easy comparison and understanding.

Types of Charts:

Chart Type	Use
Column Chart	Compare values across categories

Chart Type	Use
Pie Chart	Show proportions of a whole
Line Chart	Display trends over time
Bar Chart	Compare multiple sets of data
Area Chart	Show cumulative data

 Exercise:

Create a bar chart showing monthly sales and a pie chart showing sales distribution.

8. Conditional Formatting

Definition

Conditional formatting automatically changes the **color or style** of cells based on conditions.

Example:

Highlight marks below 35 in **red** to show failures.

Home → Conditional Formatting → Highlight Cell Rules → Less Than → 35

9. Pivot Tables and Pivot Charts (Introduction)

Definition

A **Pivot Table** summarizes large datasets quickly to analyze totals, averages, and trends.

Example:

Summarize total sales per region or per product from a large table.

Pivot Chart:

Visual representation of Pivot Table data for better understanding.

 Exercise:

Create a Pivot Table to show total marks scored by each student in all subjects.

UNIT IV – Presentation Applications (Practical)



1. Introduction to Presentation Software

Definition

A **Presentation Application** is a program used to create visual slideshows that combine text, graphics, charts, audio, and video for effective communication.

Common Tools:

- **Microsoft PowerPoint**
- **LibreOffice Impress**
- **Google Slides**
- **Canva (Online tool)**

File Extensions: .ppt, .pptx, .odp



2. Starting a Presentation

When you open PowerPoint or Impress, you can:

- Create a **New Presentation**
- Choose a **Blank Template** or **Pre-designed Theme**
- Use the **AutoContent Wizard** (provides structured templates)

Basic Elements of a Presentation:

Element	Description
Slide	A single page of the presentation
Slide Layout	Predefined arrangement of content (title, text, image, chart)
Theme	A set of coordinated colors, fonts, and effects
Background	Image or color behind the content
Placeholder	Boxes that hold text, images, or charts

3. Designing Slides

Steps for Slide Design:

1. Go to **Design Tab** → Choose a **Theme**.
2. Customize the **Background** (Solid Color, Gradient, or Image).
3. Insert **Text Boxes, Pictures, SmartArt, and Shapes**.
4. Use **Slide Master** for consistent design across all slides.

Example:

Design a presentation titled “*Digital Publishing Tools*” with 5 slides — each covering one topic like MS Word, Excel, PowerPoint, Photoshop, CorelDraw.

4. Working with Text and Objects

Formatting Text:

- Font type, size, color, alignment
 - Bullets and numbering
 - WordArt effects for titles

Working with Objects:

You can insert and format:

- **Images** (PNG, JPG)
- **Charts and Graphs** (for data representation)
 - **Tables** (for structured information)
 - **Shapes** (arrows, boxes, flowcharts)
- **SmartArt** (hierarchy, cycle, process diagrams)

Exercise:

Insert a SmartArt diagram showing “Steps in Desktop Publishing Process.”

5. Themes and Styles

Definition:

A **Theme** provides a unified look to your presentation using coordinated colors, fonts, and effects.

Style Tips:

- Use **light background** with **dark text** for readability.
 - Keep slides **clean** — avoid overcrowding.
 - Maintain **consistent fonts** across all slides.
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6. Animations and Transitions

Animations

Effects applied to objects (text, images, shapes) to make them appear or move dynamically.

Example:

- Fade In, Fly In, Zoom, Appear

Path Animation: Moves objects along a path.

Transitions

Visual effects between slides during the slideshow.

Examples:

- Push, Wipe, Dissolve, Split

Build Effects

Control the sequence in which text or objects appear on a slide (useful for presentations and demonstrations).

Exercise:

Add animation to slide headings and transitions between each slide.

7. Action Buttons and Hyperlinks

Action Buttons

Interactive buttons that perform an action during a slideshow.

Examples:

- Jump to another slide
- Open a file or website
- Play a sound or video
- End the presentation

Hyperlinks

Allow you to link slides, websites, or email addresses.

Example:

Add a hyperlink on text “*Visit Our Website*” linking to www.openai.com.



8. Rehearse Timing and Slideshow

Rehearse Timings

Used to practice and record the time spent on each slide for automatic playback.

Steps:

1. Go to **Slide Show → Rehearse Timings**
2. Adjust duration per slide
3. Save timings for automatic presentation

Slideshow Presentation

Displays the presentation in full-screen mode for delivery.

Shortcut: Press **F5** to start the slideshow.



9. Inserting Media (Sound, Video, and Charts)

Sound Effects

- Insert background music or sound for transitions and animations.

- File formats: .mp3, .wav

Video Clips

- Insert short videos related to your topic.
 - File formats: .mp4, .avi

Charts & Graphs

- Represent data visually using **Column**, **Pie**, or **Line** charts.

Exercise:

Insert a bar chart showing “Monthly Sales Report” and add a fade animation.

10. Master Slide

Definition

The **Master Slide** controls the layout, theme, and design of all slides in the presentation.

Uses:

- Add logos or watermarks to all slides.
- Maintain consistent font and color style.

Access Path:

View → Slide Master

11. Printing and Exporting Slides

- **Print Options:**

- Full Page Slides
- Notes Pages
- Handouts (multiple slides per page)

- **Export Options:**

- Save as PDF
- Save as Video (.mp4)

- Export as Images (.jpeg, .png)